AI Subseasonal to Seasonal Forecast Guidance for India

Experimental Research

Initialization: 8 November 2026, 00 UTC | **Forecast period:** 8 November–5 December 2026

Nov 9, 2026 ·

Executive Summary

The 8 November initialization indicates a dry southeastern peninsula alongside a wet Northeast India, eastern Bay and equatorial Indian Ocean in Week 1, small departures over land in Week 2, and a southern dry spell that deepens through Weeks 3–4. In Week 1, Tamil Nadu, coastal Andhra Pradesh and Rayalaseema are drier than normal, while Mizoram and Manipur are wetter than normal and the Andaman and Nicobar Islands are +10 to +15 mm/day. Week 2 (15–21 November) has only a weak wet signal over the central Bay and the south Andhra Pradesh coast. In Weeks 3–4, Tamil Nadu, Kerala and the southwest Bay turn increasingly dry, reaching −5 to −10 mm/day over Tamil Nadu in Week 4.

Temperature anomalies are split between a warm northwest and a cool east. Punjab, Haryana, Delhi and Rajasthan are +2 to +4°C in Week 1, and the western Himalaya is warmer still, while Northeast India, Sikkim and the eastern peninsula are cooler than normal. The India-mean departure falls from about +0.9°C in Week 1 to near zero in Week 4 as the cool area spreads.

The 850-hPa circulation has cyclonic anomalies over the southern Bay near 7.5–10.5°N, 88–90°E in Week 1, beside the wet band there. In Week 2 speeds are reduced over Kerala, Tamil Nadu and the southwest Bay. Weeks 3–4 are dominated by anticyclonic anomalies over the southern Bay, the southern peninsula and the southeast Arabian Sea, matching the deepening southern dryness.

Cyclogenesis indices (GPI, KGPP, IGPP*) are included below.

1 Methodology

The writeup interprets the rainfall, 2-m temperature and 850-hPa wind-anomaly maps of the AI S2S forecast initialized at 00 UTC on 8 November 2026. It covers four consecutive seven-day periods: 8–14 November, 15–21 November, 22–28 November and 29 November–5 December 2026. The forecast is an AI-based multi-model ensemble combining several AI and physics-based S2S forecast systems. Anomalies are against a seasonally matched climatology. Maps use visual bilinear interpolation from the native 1.5° grid to 0.05°.

Rainfall departures are in mm/day, temperature departures in °C, and wind-speed departures in m/s. Purple rainfall shading denotes wetter-than-normal conditions; yellow and orange denote drier-than-normal conditions. The white rainfall interval is −2 to +2 mm/day, so white areas can still receive rain. Temperature shading is asymmetric around zero: pale yellow represents 0 to +1°C, the palest lavender −1 to 0°C, and darker purple larger negative departures; orange to red mark departures of +1°C and above. In the wind maps, shading is the ensemble-mean vector speed minus the climatological mean speed: blue indicates greater-than-normal speed and brown lower-than-normal speed, with near-white spanning about −1 to +1 m/s. Streamlines show the U/V component anomaly. The direction of that anomaly is distinct from the total forecast wind, and a positive speed anomaly does not by

itself establish a stronger flow in any particular direction. Cyclonic and anticyclonic centres named below were checked against the relative vorticity of the U/V anomaly on the native grid.

Ranges and boundaries quoted below are readings of the plotted colors, spot-checked against state and box averages of the plotted field and native-grid extremes. No district-level totals or calibrated probabilities are inferred.

2 Rainfall Forecast

Week 1 — 8–14 November

A dry anomaly covers the southeastern peninsula. Tamil Nadu, coastal Andhra Pradesh and Rayalaseema are −2 to −5 mm/day, reaching −5 to −10 mm/day along the north Tamil Nadu–south Andhra Pradesh coast near 12–15°N. Kerala and Karnataka are slightly dry, reaching −2 to −3 mm/day in places.

Above-normal rainfall covers Northeast India, with +2 to +5 mm/day over Mizoram and Manipur; Tripura is slightly wet. A broad wet band of +5 to +10 mm/day runs along the eastern Bay from the Myanmar coast to the Andaman Sea, with +10 to +15 mm/day over the Andaman and Nicobar Islands. The equatorial Indian Ocean south of about 8°N is strongly wet, at +10 to +20 mm/day near 0–6°N and 85–92°E. Central and northern India have small departures.

Week 2 — 15–21 November

Departures over most of India fall within the small-anomaly interval. A weak wet signal of +2 to +5 mm/day lies over the central Bay near 11–16°N and 82–90°E and over the north Bay along the Myanmar coast; over land the south coastal Andhra Pradesh and Telangana are only slightly wetter than normal. Mizoram has a small wet patch.

The equatorial Indian Ocean turns dry, at −2 to −5 mm/day south of about 8°N and locally −5 to −10 mm/day near 2–4°N, 70–75°E and off northern Sumatra. Northern and northwestern India have small departures.

Week 3 — 22–28 November

The southern dry signal develops. Tamil Nadu is −2 to −5 mm/day, locally −5 mm/day or lower, and Kerala and south coastal Andhra Pradesh are −2 to −5 mm/day. The southwest Bay is −2 to −5 mm/day, reaching −5 to −10 mm/day near 6–12°N and 80–87°E.

A dry band of −5 to −10 mm/day covers the equatorial Indian Ocean south of Sri Lanka and India. The rest of India has small departures.

Week 4 — 29 November–5 December

The southern dry signal is strongest. Tamil Nadu is −5 to −10 mm/day, Kerala −2 to −5 mm/day, and Andhra Pradesh and Rayalaseema −2 to −5 mm/day, locally −5 to −10 mm/day along the south Andhra Pradesh coast. The southwest Bay is −5 to −10 mm/day, reaching about −10 mm/day near 12°N, 81°E, and the central Bay, the Andaman Sea and the southeast Arabian Sea are −2 to −5 mm/day.

Jammu & Kashmir has a small wet patch. A weak wet band of +2 to +5 mm/day lies over the equatorial Indian Ocean near 3–8°N and 60–75°E.

Main rainfall interpretation: Week 1 is dry over the southeastern peninsula and wet over Northeast India, the eastern Bay and the equatorial ocean. Week 2 is close to normal over

land. Weeks 3–4 bring a deepening dry spell over Tamil Nadu, Kerala, south coastal Andhra Pradesh and the southwest Bay.

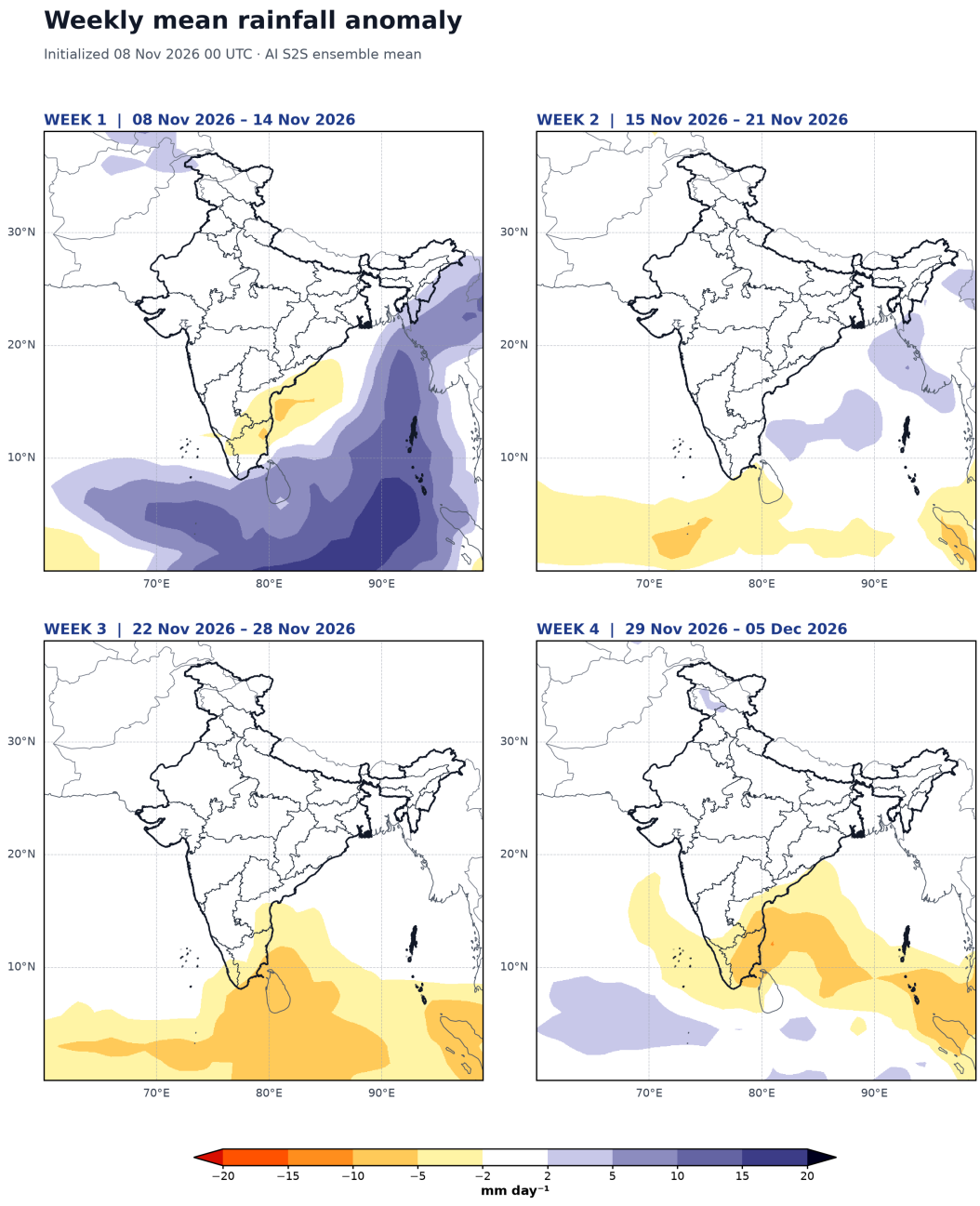


Figure 1. Weekly mean rainfall anomalies, 8 November–5 December 2026 (mm/day), AI S2S ensemble mean. Purple indicates above-normal rainfall; yellow/orange indicates below-normal rainfall. White spans −2 to +2 mm/day.

3 Temperature Forecast

Week 1 — 8–14 November

Warmth concentrates in the northwest and the western Himalaya. Jammu & Kashmir, Ladakh and Himachal Pradesh are +3 to +5°C, locally higher over high terrain, and Punjab, Rajasthan, Haryana and Delhi are +2 to +4°C. Gujarat and Uttar Pradesh are +1 to +2°C.

Negative departures cover Northeast India and Sikkim, at −1 to −3°C over Sikkim and 0 to −2°C over Assam and Mizoram. A cool strip of 0 to −2°C covers Telangana, Chhattisgarh, Odisha and coastal Andhra Pradesh. Kerala and Karnataka are mostly 0 to +1°C.

Week 2 — 15–21 November

Warmth stays in the west and northwest: Rajasthan and Gujarat are +1 to +3°C, Punjab, Haryana and Delhi +1 to +2°C, and the western Himalaya +2 to +5°C. Madhya Pradesh and Uttar Pradesh are +1 to +2°C.

The cool area widens. Assam, Mizoram and Sikkim are −1 to −3°C, and Andhra Pradesh, Telangana, Odisha, Tamil Nadu and Chhattisgarh are 0 to −2°C.

Week 3 — 22–28 November

Warmth retreats to the west: Gujarat and Rajasthan are +1 to +2°C, and Punjab, Haryana and Delhi 0 to +1°C. Jammu & Kashmir and Ladakh are +1 to +4°C.

Cooler-than-normal conditions cover the east and the southeast peninsula: Odisha, Telangana, Tamil Nadu, Andhra Pradesh, Chhattisgarh and Kerala are 0 to −2°C, and Assam, Mizoram and Sikkim −1 to −3°C.

Week 4 — 29 November–5 December

Gujarat and Rajasthan remain +1 to +2°C, and Punjab, Haryana and Delhi 0 to +1°C. Jammu & Kashmir and Ladakh are +1 to +4°C.

The cool area reaches its widest extent. Odisha, Telangana, Chhattisgarh, Andhra Pradesh, Tamil Nadu, Bihar, Jharkhand and West Bengal are 0 to −2°C, mostly −1 to −2°C over Odisha and Telangana, and Assam and Sikkim are −2 to −3°C. These are weekly mean 2-m temperature anomalies and do not specify separate maximum or minimum temperature forecasts.

Main temperature interpretation: The northwest and the western Himalaya are warmer than normal throughout, most strongly in Weeks 1–2, while Northeast India and the eastern peninsula are cooler than normal and the cool area spreads with time. The India-mean departure falls from about +0.9°C in Week 1 to near zero in Week 4.

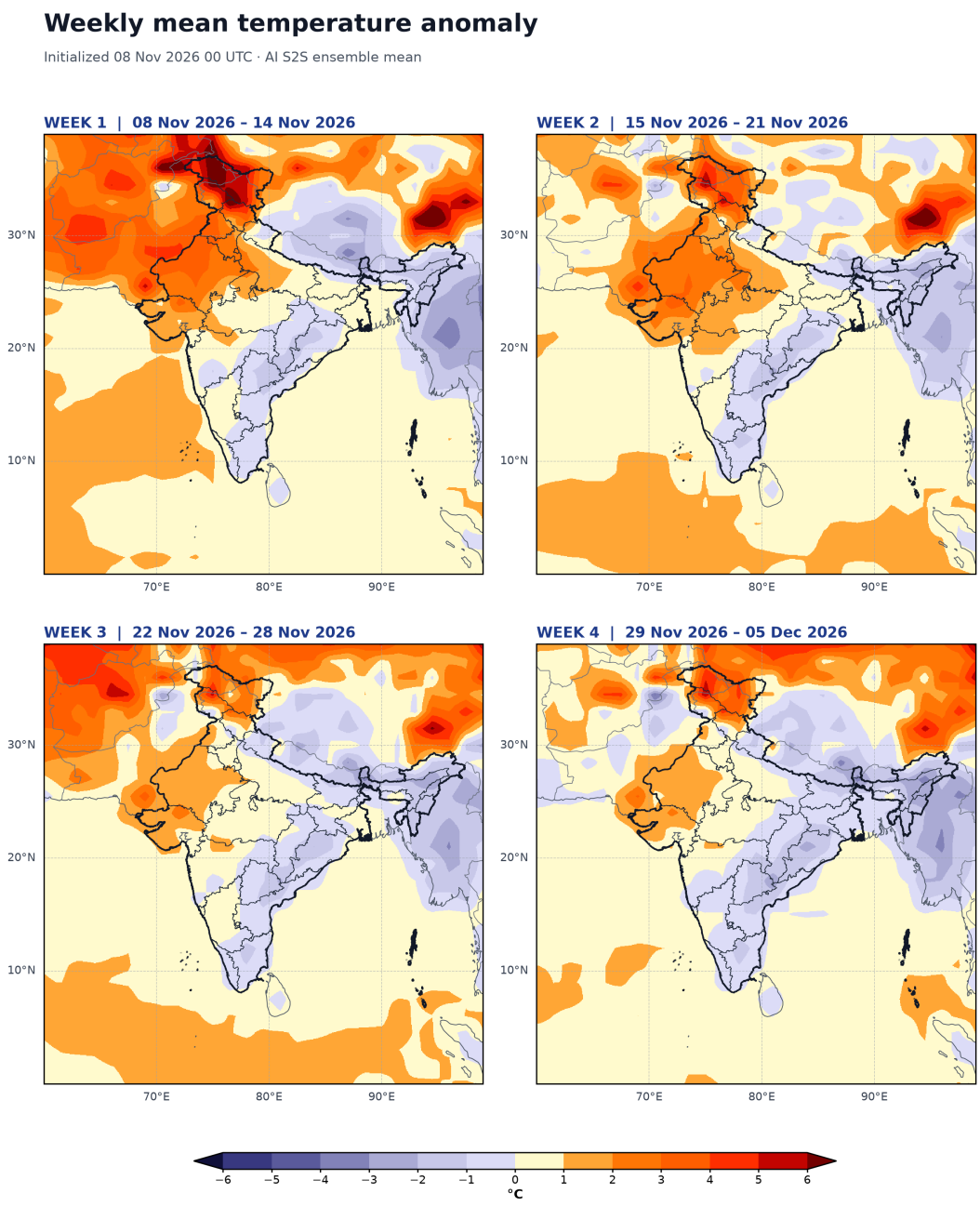


Figure 2. Weekly mean 2-m temperature anomalies, 8 November–5 December 2026 (°C), AI S2S ensemble mean. Purple indicates cooler-than-normal conditions; yellow/orange/red indicates warmer conditions. Pale yellow represents 0 to +1°C.

4 Low-Level Wind and Circulation Forecast

Week 1 — 8–14 November

Cyclonic anomalies lie over the southern Bay near 7.5°N, 90°E and 10.5°N, 88.5°E, beside the Week 1 wet band. Speeds are enhanced by +2 to +4 m/s over the eastern Bay and the

Andaman Sea and by +1 to +3 m/s across the Arabian Sea toward the Gujarat–Konkan coast, with an anticyclonic anomaly near 22.5°N, 72°E.

Speeds are reduced by −1 to −2 m/s over West Bengal, Bihar, Jharkhand, Odisha and eastern Uttar Pradesh.

Week 2 — 15–21 November

Speeds are reduced by −2 to −3 m/s over Kerala and Tamil Nadu and by −2 to −4 m/s over the southwest Bay, with westerly anomalies across the southern peninsula and a weak cyclonic centre over the southwest Bay near 10.5°N, 84°E. A cyclonic anomaly lies over the north Bay near 19.5°N, 91.5°E.

Speeds are enhanced by +1 to +3 m/s over Uttar Pradesh, Delhi, Haryana and Punjab, and reduced by −1 to −2 m/s over Gujarat. These weekly anomaly patterns do not identify a surface depression; a cyclone assessment would require the total circulation, surface pressure and other fields.

Week 3 — 22–28 November

Anticyclonic anomalies lie over the southern Bay near 6°N, 82.5°E, over the southern peninsula near 13.5°N, 79.5°E and over the southeast Arabian Sea near 9°N, 75°E. Speeds are reduced by −1 to −2 m/s over Telangana, Andhra Pradesh, Maharashtra, Gujarat and Karnataka, and by −2 to −3 m/s over the southwest and central Bay and the Andaman Sea. Easterly anomalies with strongly reduced speeds of −2 to −8 m/s cover the equatorial ocean south of about 5°N.

Week 4 — 29 November–5 December

Anticyclonic anomalies cover the southern peninsula and the southeast Arabian Sea, with centres near 12°N, 79.5°E, 7.5°N, 75°E and 15°N, 67.5°E. This coincides with the Week 4 southern dry signal. Speeds are reduced by −1 to −3 m/s over Gujarat, Karnataka, Andhra Pradesh and Maharashtra, and by −1 to −2 m/s over the central Bay and the Andaman Sea. A small area of enhanced speed lies over the Gulf of Mannar near 7–9°N.

Main circulation interpretation: Cyclonic anomalies over the southern Bay in Week 1 accompany the wet band there. From Week 2 onward speeds are reduced over the southern peninsula and the southwest Bay, and Weeks 3–4 bring anticyclonic anomalies over the southern Bay, the southern peninsula and the southeast Arabian Sea, matching the deepening southern dryness. Directional change and wind-speed change are distinct signals.

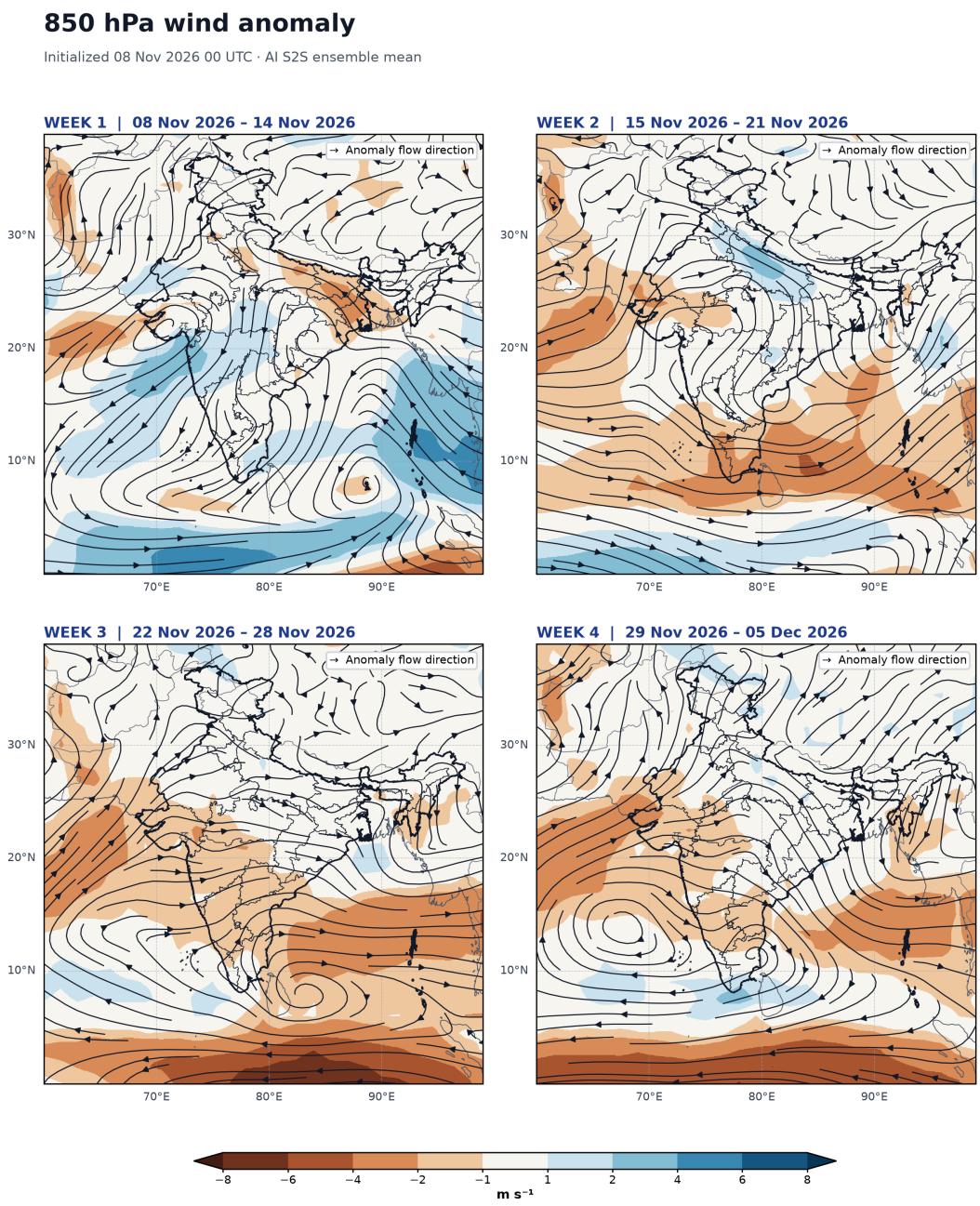


Figure 3. Weekly 850-hPa wind-speed anomalies and U/V anomaly streamlines, 8 November–5 December 2026, AI S2S ensemble mean. Blue denotes stronger-than-normal speed; brown denotes weaker speed. Arrowheads indicate anomalous-flow direction, not total forecast wind.

5 Key S2S Signals

- **Dry southeastern peninsula in Week 1:** Tamil Nadu, coastal Andhra Pradesh and Rayalaseema are −2 to −5 mm/day, with −5 to −10 mm/day along the north Tamil Nadu–south Andhra coast.

- **Wet northeast, eastern Bay and equatorial ocean in Week 1:** Mizoram and Manipur are +2 to +5 mm/day, the Andaman and Nicobar Islands +10 to +15 mm/day, and the equatorial Indian Ocean up to +10 to +20 mm/day, with cyclonic anomalies over the southern Bay.
- **Near-normal Week 2 over land (15–21 November):** only a weak wet signal over the central Bay and the south Andhra Pradesh coast.
- **Deepening southern dry spell in Weeks 3–4:** Tamil Nadu −2 to −5 mm/day in Week 3 and −5 to −10 mm/day in Week 4, with Kerala, south coastal Andhra Pradesh and the southwest Bay also dry, under anticyclonic anomalies.
- **Warm northwest, cool east:** +2 to +4°C over Punjab, Haryana, Delhi and Rajasthan in Week 1 and a warm western Himalaya throughout, against a cool Northeast India and eastern peninsula that spreads with time; India-mean +0.9°C in Week 1, near zero by Week 4.

6 Interpretation

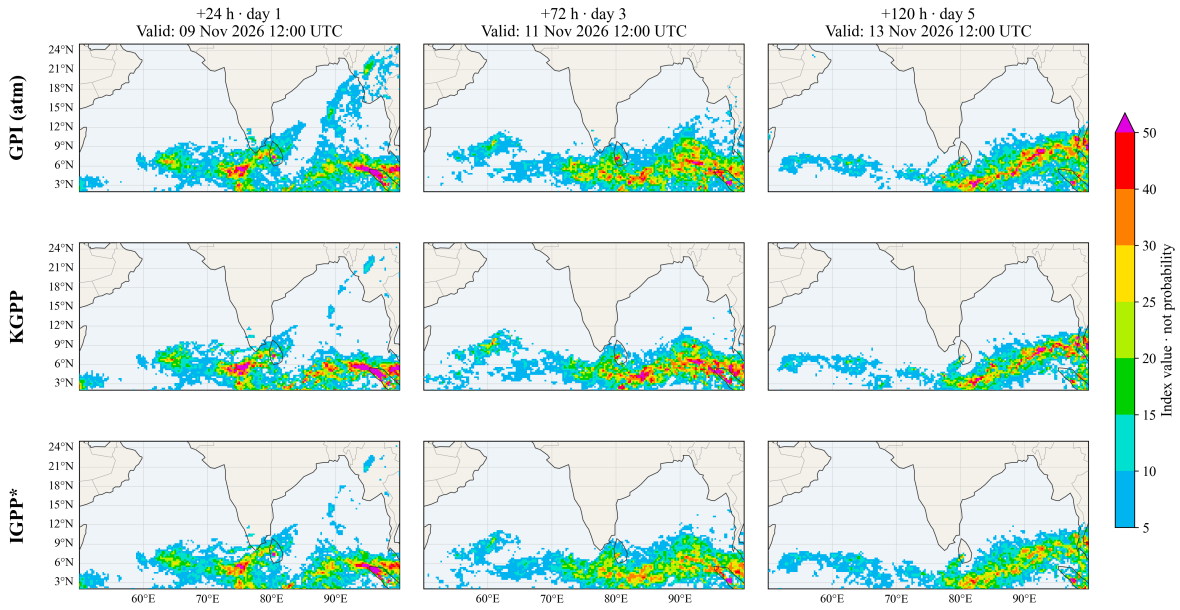
The most coherent evolution is a dry southeastern peninsula and a wet eastern Bay in Week 1, a near-normal Week 2 over land, and a southern dry spell that deepens through Weeks 3–4. The wind fields support each stage: cyclonic anomalies over the southern Bay accompany the Week 1 wet band, and anticyclonic anomalies over the southern Bay, the southern peninsula and the southeast Arabian Sea accompany the Week 3–4 dryness. These weekly means do not establish the daily sequence of individual rain events.

The forecast period falls in the northeast monsoon season, when Tamil Nadu, coastal Andhra Pradesh, Rayalaseema and Kerala receive much of their annual rainfall. Dry conditions over Tamil Nadu in Weeks 1, 3 and 4, with no compensating wet week, would mean below-normal northeast monsoon rainfall there over the four weeks. The supplied maps do not show the total wind, so they do not establish a monsoon onset or break.

October–December is an important North Indian Ocean cyclone season. The Week 1 cyclonic anomalies over the southern Bay, where the AI cyclogenesis indices are also elevated over 9–13 November, merit monitoring in subsequent runs. A weekly-mean anomaly circulation is not evidence of a depression or cyclone. [Pattanaik & Mohapatra (2021), MAUSAM]

Overall, the 8 November initialization supports a dry start over the southeastern peninsula with rain over Northeast India and the eastern Bay, near-normal rainfall over land in mid-November, a deepening dry spell over Tamil Nadu, Kerala and the southwest Bay from late November into early December, and a warm northwest against a cool east.

Cyclogenesis



Sampled eight-member ensemble mean; one fixed initialization, columns show future valid times. Native 0.25° fields.
GPI (atm) omits potential intensity; IGPP* uses the report's 0.1 shear scaling. Research indices, not IMD warnings.
Values below 5 (and invalid cells) are unshaded; magenta is ≥50. Not cyclone probabilities.

Above are initialized at 12:00 UTC on 8 November 2026, the newest AI cyclogenesis forecast run published when this guidance was prepared, and show the mean of the first 8 of its 64 ensemble members. The three forecast columns are valid at 12:00 UTC on 9, 11 and 13 November (+24, +72 and +120 hours).

At +24 hours, elevated index values form a band along about 3–9°N from the central Arabian Sea, south of India and Sri Lanka, to the Andaman Sea and northern Sumatra. All three rows have magenta cores in two places: over the sea southwest of the Indian tip near 5.5°N, 75.5°E, where KGPP reaches about 120 and GPI about 80, and at the northern tip of Sumatra near 5°N, 96°E, where GPI and KGPP reach about 150, largely over land. GPI and IGPP* also saturate over Sri Lanka itself. A sparse patch of low values extends up the eastern Bay toward the Myanmar coast, but north of 10°N no index exceeds about 26.

At +72 hours, the Arabian Sea part of the band weakens: no Arabian Sea cell reaches 50, and a separate patch of up to about 30–40 sits near 9–10°N, 60–61°E. The strongest colors shift to the southern Bay and the equatorial belt east of about 85°E, with KGPP up to about 85 near 5.5°N, 98°E and isolated GPI cores of about 60 near 6°N, 93°E; the largest values, above 100, lie over land in northern Sumatra. IGPP* is more restrained, with no Bay of Bengal cell at or above 50.

At +120 hours, the Arabian Sea is nearly blank above 25, and the colored area concentrates in a band running from south of Sri Lanka (GPI about 90 near 3°N, 82°E) northeastward across the southern and eastern Bay to the Andaman Sea. KGPP peaks at about 85 near 7.75°N, 92°E, and GPI reaches about 60–75 near 9–10°N, 98–100°E at the eastern edge. Over the three leads the favorable environment therefore moves from the southern Arabian Sea and the waters south of India toward the southern and eastern Bay of Bengal, while staying south of about 10°N. The lowest closed sea-level-pressure low in any of the 8 members between +24 and +120 hours is 1008 hPa and no more than 2.5 hPa below its surrounding ring, so the members show no organized depression in this period.

The figure does not show member spread, a storm track, a closed circulation, a warm core or observed genesis. Different rows use different formulas, so their colored areas should not be read as agreement on a probability. Magenta means an index value of at least 50, where the color scale saturates. Blank cells may be below 5 or invalid, and they do not establish an absence of hazardous weather. High values can also occur over land, as over Sri Lanka and Sumatra here. This is a research view of environmental favorability, not an IMD warning or a forecast of a specific cyclone's formation.

Data and Product Notes

Figures 1–3 are maps produced for this issue from the weekly AI S2S ensemble-mean fields on the native 1.5° grid, drawn with the colour scales and layout of s2s.scdlds.com; they are not screenshots of it. State-level coverage and ranges are readings of those maps, cross-checked against state and box averages of the plotted fields; they are not official regional statistics.

Cyclogenesis indices (GPI, KGPP, IGPP*) are from the AI cyclogenesis forecast initialized at 12 UTC on 8 November 2026 (mean of 8 sampled members). Web sources establish monsoon and cyclone-season context; they are not used as verification of this experimental forecast.

Forecast product website: S2S experimental forecast products — s2s.scdlds.com

Scientific and official references

- IMD extended-range cyclone guidance and post-monsoon season context. [Pattanaik & Mohapatra (2021), MAUSAM]
- Cyclogenesis index recipe (GPI, KGPP, IGPP). [Ganesh et al. (2021), MAUSAM, https://doi.org/10.54302/mausam.v72i1.126]
- Indian meteorological subdivision names and geographic context. [IMD, meteorological subdivision map and legend]

This document is experimental research guidance and is not an operational warning.